

Section 2: Conditional Probability

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1 Introduction

1.1 Logistics

Welcome! The goal of our weekly section is to review content from lecture, with emphasis on big-picture intuition, mathematical derivation, and hands-on practice.

Specifically, we aim for section to be interactive, well-paced, inclusive, and fun! Please do not hesitate to reach out if you have any questions/feedback!

1.2 Office Hours

- See the Google Calendar for the most accurate information!

1.3 Attendance

- bit.ly/stat110harvard

2 Non-Naive Probability

Idea. We'll now generalize probability to the *non-naive case*, which allows for much more flexibility. As we'll see, all the important properties of probability can be built from simply two axioms.

Definition 1 (Axioms of probability). A *probability space* consists of a *sample space* S and a *probability function* P , which takes an *event* $A \subseteq S$ as input and returns $P(A)$, a real number between 0 and 1, as output. The function P must satisfy the following axioms:

$$P(\emptyset) = 0,$$

$$P(S) = 1,$$

and if $A_1, A_2, \dots$ are disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Definition 2 (Properties of probability). Informally, *probability* is a measure of how likely an event is, bounded from 0 to 1.¹ With the axioms above, probability satisfies the following properties:

$$P(A^c) = 1 - P(A),$$

¹In the *Frequentist* paradigm, probability can be viewed as long-term frequency. E.g., the probability of an ace is $p = \frac{1}{13}$, so in the long run, we expect 1 ace for every 13 cards.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

and

$$A \subseteq B \implies P(A) \leq P(B).$$

- 💡: The complement is useful for finding the event of “at least” something.

✎ Concept Checker 1. If I roll a fair 6-sided die three times, what is the probability that I land 5 or higher at least once?

Solution

Let A be the event that I land 5 or higher at least once. We want $P(A)$, but since this is the event of “at least” something, it’s easier to find the complement. (Notice I can roll 5 on only the first roll, 5 on only the second roll, 5 on the first two rolls, and so on, so the situation quickly gets complicated!)

$$\begin{aligned} P(A) &= 1 - P(A^c) && \text{by complement rule} \\ &= 1 - (4/6)^3 && \text{since we can roll 1, 2, 3, or 4} \\ &= 19/27 && \text{by simplifying} \end{aligned}$$

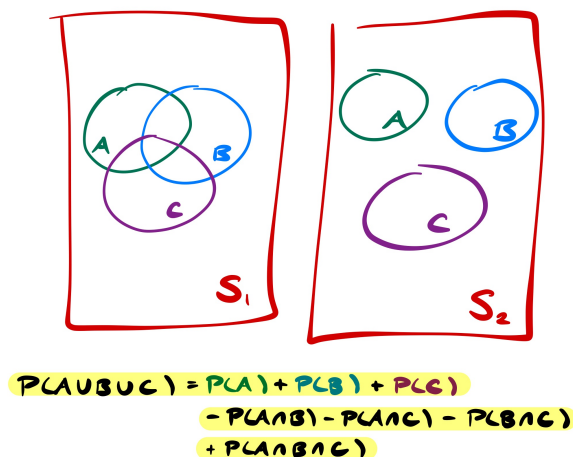


FIGURE 1: The sample space with Venn diagrams illustrating PIE for 3 events.

Definition 3 (Principle of inclusion-exclusion (PIE)). A useful tool for finding the probability of *unions* of events.

- Informally,

$$P(\text{union of many events}) = \text{singles} - \text{doubles} + \text{triples} - \text{quadruples} + \dots$$

- Formally,

$$P(A_1 \cup \dots \cup A_n) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n).$$

- E.g., with 3 events,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

Definition 4 (Symmetry). Some problems can be greatly simplified if they contain *symmetry* such that *outcomes are equally likely*.

- E.g., if $P(A_1) = P(A_i)$ for all i , $P(A_1 \cap A_2) = P(A_i \cap A_j)$ for all $i \neq j$, and so on, then PIE simplifies to $P(A_1 \cup \dots \cup A_n) = \binom{n}{1}P(A_1) - \binom{n}{2}P(A_1 \cap A_2) + \binom{n}{3}P(A_1 \cap A_2 \cap A_3) - \dots + (-1)^{n+1}P(A_1 \cap \dots \cap A_n)$.

 **Concept Checker 2.** With 3 events, what is $P(A \cup B \cup C)$ if there is no overlap between A , B , and C (i.e., if $P(A \cap B) = P(B \cap C) = P(A \cap C) = P(A \cap B \cap C) = 0$)?

Solution

If there is no overlap between A , B , and C , then $P(A \cup B \cup C) = P(A) + P(B) + P(C)$ (i.e., the sum of the marginal probabilities). PIE adds/subtracts additional terms to adjust for overlap, but in this simple case, there is none!

3 Joint, Marginal, and Conditional Probabilities

Idea. We'll define some important terms to introduce the idea of *conditional probability*, one of the most important concepts in all of statistics. With it, we'll expand our toolkit to calculate various types of probabilities (conditional, joint, marginal, etc.).

Definition 5 (Joint probability). $P(A \cap B) = P(A, B)$ is the *joint probability* of A and B .

- Importantly, $P(A \cap B) = P(B \cap A)$.
- 💡: We can treat $A \cap B$ (or any union or intersection) as *one* event! Thus, all the previous rules carry over. E.g., $P((A \cap B)^c) = 1 - P(A \cap B)$ by complement rule.

Definition 6 (Marginal probability). $P(A)$ is the *marginal probability* of A (i.e., “by itself”).

- Equivalently, this is the *unconditional* or *prior* probability.

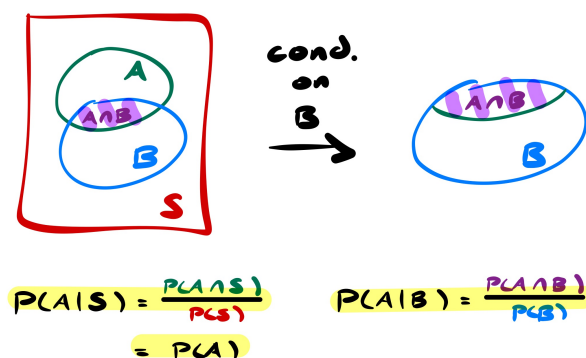


FIGURE 2: The sample space before and after conditioning.

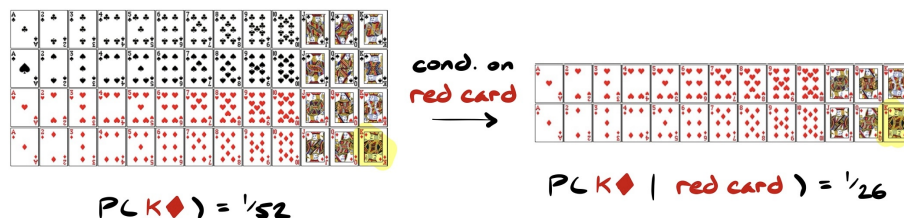


FIGURE 3: A standard deck of 52 playing cards shrunk down to only the 26 red cards.

Definition 7 (Conditional probability).

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

is the *conditional probability* of A given that B occurred, where $P(B) > 0$.

- When we *condition* on B , we reduce the sample space to the area where B occurs (i.e., we divide by $P(B)$). Within this new space, the area where A occurs is the *intersection* of A and B (i.e., we calculate $P(A \cap B)$). This is illustrated in Figure 2 and Figure 3.
- ☞: In general, $P(A | B) \neq P(B | A)$!
- 💡: All conditional probabilities *are* probabilities, and all probabilities *are* conditional probabilities (implicitly conditioning on S). Thus, all the previous rules carry over. E.g., $P(A^c | B) = 1 - P(A | B)$ by complement rule.

✎ **Concept Checker 3.** Rearrange the definition of conditional probability to get two formulas for $P(A \cap B)$. How do these connect to the multiplication rule?

Solution

$P(A | B) = \frac{P(A \cap B)}{P(B)} \implies P(A \cap B) = P(B)P(A | B)$. $P(B | A) = \frac{P(B \cap A)}{P(A)} \implies P(B \cap A) = P(A \cap B) = P(A)P(B | A)$. Using the same logic as the multiplication rule, we can imagine two tree diagrams. First, A happens, and then B happens (given that A happened). Equivalently, B happens, and then A happens (given that B happened).

✎ **Concept Checker 4.** Using the multiplication rule, what is a formula for $P(A_1, \dots, A_n)$? How many formulas are there in total?

Solution

$P(A_1, \dots, A_n) = P(A_1)P(A_2 | A_1)P(A_3 | A_1, A_2) \cdots P(A_n | A_1, \dots, A_{n-1})$. There are $n!$ total formulas (since we could've started with $P(A_i)$ for any of the n available i , followed by $P(A_j | A_i)$ for any of the $n - 1$ remaining j , and so on).

Definition 8 (Bayes' Rule).

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)},$$

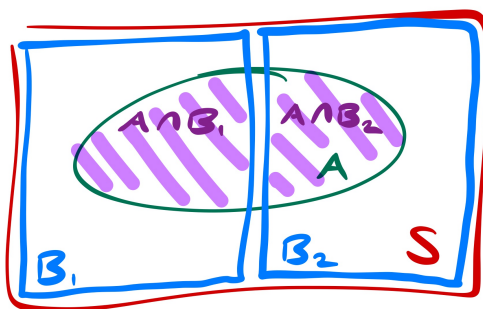
where $P(B) > 0$.

- 💡: Often, we want to find $P(A | B)$ but are given $P(B | A)$. This is where Bayes' Rule is useful!

✎ **Concept Checker 5.** Derive Bayes' Rule using the definition of conditional probability.

Solution

$$\begin{aligned} P(A | B) &= \frac{P(A \cap B)}{P(B)} && \text{by definition of conditional probability} \\ &= \frac{P(A)P(B | A)}{P(B)} && \text{since } P(A \cap B) = P(A)P(B | A) \end{aligned}$$



$$\begin{aligned} P(A) &= P(A \cap B_1) + P(A \cap B_2) \\ &= P(A | B_1)P(B_1) + P(A | B_2)P(B_2) \end{aligned}$$

FIGURE 4: The sample space partitioned by B_1 and B_2 illustrating LOTP for 2 events.

Definition 9 (Partition). Events $B_1, \dots, B_n$ partition the sample space if each pairwise intersection is empty and their union is the entire sample space.

- Mathematically, $B_i \cap B_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^n B_i = S$ such that $\sum_{i=1}^n P(B_i) = 1$.

Definition 10 (Law of Total Probability (LOTP)).

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i),$$

where $B_1, \dots, B_n$ partition the sample space.

- Since $B_1, \dots, B_n$ partition the sample space, $P(A) = P(A \cap B_1) + \dots + P(A \cap B_n)$, and $P(A \cap B_i) = P(A | B_i)P(B_i)$. This is illustrated in Figure 4.
- 💡: LOTP is useful when we want to condition on something (i.e., “wishful thinking”). Also, LOTP is useful to find the denominator in Bayes' Rule.

✎ **Concept Checker 6.** Suppose that every upperclassman has a probability p of owning a scooter. If they live in the Quad, $p = \frac{1}{10}$. Otherwise, $p = \frac{1}{20}$. What is the probability that a randomly selected upperclassman owns a scooter?

Solution

Let S be the event that they own a scooter and Q be the event that they're in the Quad. Notice the wishful thinking, where we really want to know whether they're in the Quad or not.

$$\begin{aligned}
 P(S) &= P(S \mid Q)P(Q) + P(S \mid Q^c)P(Q^c) && \text{by LOTP} \\
 &= (1/10)(3/12) + (1/20)(9/12) && \text{since } p = \frac{1}{10} \text{ for the 3 Quad houses} \\
 &= 0.0625 && \text{by simplifying}
 \end{aligned}$$

Notice $P(S) = 0.0625$ is weighted toward $p = \frac{1}{20}$, which makes sense since the majority of students don't live in the Quad and thus have a lower probability of owning a scooter.

4 Disjoint and Independent Events

Idea. Different events are often related to each other, whether by being *disjoint* or *independent*. Importantly, things get more complicated with three (or more) events.

Definition 11 (Disjointness). Events A and B are *disjoint* if they cannot happen simultaneously.

- Mathematically, $P(A \cap B) = 0$.

Definition 12 (Independence). Events A and B are *independent*, denoted $A \perp\!\!\!\perp B$, if knowing one gives no information about the other.

- Mathematically, $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$, which imply $P(A \cap B) = P(A)P(B)$.
- If $A \perp\!\!\!\perp B$, then $A^c \perp\!\!\!\perp B$, $A^c \perp\!\!\!\perp B^c$, and $A \perp\!\!\!\perp B^c$.

Definition 13 (Independence of three events). Events A , B , and C are *independent* if $P(A \cap B) = P(A)P(B)$, $P(A \cap C) = P(A)P(C)$, $P(B \cap C) = P(B)P(C)$, and $P(A \cap B \cap C) = P(A)P(B)P(C)$.

- If only the first three conditions hold, then A , B , and C are *pairwise independent*, but the last condition is needed for A , B , and C to be *independent*.

Definition 14 (Conditional independence). Events A and B are *conditionally independent given C* , denoted $(A \perp\!\!\!\perp B) \mid C$, if, conditional on C , knowing one gives no information about the other.

- Mathematically, $P(A \mid B, C) = P(A \mid C)$ and $P(B \mid A, C) = P(B \mid C)$, which imply $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$. Notice this takes the definition of (unconditional) independence and adds “ $\mid C$ ” to everything.
- $\otimes$: Conditional independence does *not* imply (unconditional) independence, and vice versa.

Example 1 (Unconditional independence). I have two fair 6-sided dice, so the rolls are *independent* (e.g., if I roll a 6 on the first one, I still don't know what I'll get on the next

one). But if we condition on the event that the sum is 10, the second roll is deterministic from the first one and thus *not conditionally independent*.

- Mathematically, let X_i be the value of my i th roll. $\{X_1 = 6\} \perp\!\!\!\perp \{X_2 = 4\}$, but $(\{X_1 = 6\} \not\perp\!\!\!\perp \{X_2 = 4\}) \mid \{X_1 + X_2 = 10\}$.

Example 2 (Conditional independence). Let p be the probability I make a free throw. If I shoot 9 free throws that all miss, you'd probably assume that the 10th shot would miss because the event that I miss 9 shots implies that p is low. But if we condition on $p = 0.5$, then the 10th shot is *conditionally independent* of the previous 9; I'm just having an unlucky day.

- Mathematically, let A be the event that I miss the first 9 shots and B be the event that I miss the 10th shot. $A \not\perp\!\!\!\perp B$, but $(A \perp\!\!\!\perp B) \mid \{p = 0.5\}$.

5 Extensions of Bayes' Rule and LOTP

Idea. When working with three events, we can extend our previous tools using two main tricks: extra conditioning, and treating an intersection as one event.

Definition 15 (Bayes' Rule with extra conditioning).

$$P(A \mid B, C) = \frac{P(B \mid A, C)P(A \mid C)}{P(B \mid C)},$$

where $P(B \mid C) > 0$.

- Notice that this takes “regular” Bayes' Rule—i.e., $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$ —and adds “ $\mid C$ ” to everything.

Definition 16 (Bayes' Rule with B, C as one event).

$$P(A \mid B, C) = \frac{P(B, C \mid A)P(A)}{P(B, C)},$$

where $P(B, C) > 0$.

- Notice that this takes “regular” Bayes' Rule—i.e., $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$ —and replaces B with B, C .

Definition 17 (LOTP with extra conditioning).

$$P(A \mid C) = \sum_{i=1}^n P(A \mid B_i, C)P(B_i \mid C),$$

where $B_1, \dots, B_n$ partition the sample space.

- Notice that this takes “regular” LOTP—i.e., $P(A) = \sum_{i=1}^n P(A \mid B_i)P(B_i)$ —and adds “ $\mid C$ ” to everything.

 **Concept Checker 7.** Bayes' Rule with extra conditioning requires that $P(B \mid C) > 0$, and Bayes' Rule with B, C as one event requires that $P(B, C) > 0$. Why are these the same condition?

Solution

$P(B | C) > 0 \implies P(B, C) > 0$ since $P(B | C) = \frac{P(B, C)}{P(C)}$, and $P(B, C) > 0 \implies P(B | C) > 0$ since $P(B, C) = P(C)P(B | C)$. Thus, $P(B | C) > 0 \iff P(B, C) > 0$.

6 Recap

Idea. Let's review the main rules and strategies we've discussed for calculating probability.

Definition 18 (Probability toolkit). Informally, *probability* is a measure of how likely an event is, bounded from 0 to 1.

- $P(A) = \frac{\# \text{ of outcomes favorable to } A}{\# \text{ of total outcomes in } S} = \frac{|A|}{|S|}$ by Naive Probability, where the sample space is finite and consists of equally likely outcomes.
- $P(A) \leq P(B)$ if A implies B .
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ by PIE.
- $P(A \cap B) = P(A, B) = P(A)P(B | A) = P(B)P(A | B)$ by Multiplication Rule.
- $P(A) = 1 - P(A^c)$ by Complement Rule.
- $P((A \cup B)^c) = P(A^c \cap B^c)$, $P((A \cap B)^c) = P(A^c \cup B^c)$ by De Morgan's Laws.
- $P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A | B_i)P(B_i)$ by LOTP, where $B_1, \dots, B_n$ partition the sample space.
- $P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$ by Bayes' Rule.
- $P(A | C) = \sum_{i=1}^n P(A \cap B_i | C) = \sum_{i=1}^n P(A | B_i, C)P(B_i | C)$ by LOTP with Extra Conditioning, where $B_1, \dots, B_n$ partition the sample space.
- $P(A | B, C) = \frac{P(A \cap B | C)}{P(B | C)} = \frac{P(B|A, C)P(A | C)}{P(B | C)}$ by Bayes' Rule with Extra Conditioning.
- $P(A | B, C) = \frac{P(A \cap B, C)}{P(B, C)} = \frac{P(B, C | A)P(A)}{P(B, C)}$ by Bayes' Rule with B, C as One Event.
- 💡: The complement is useful for finding the event of “at least” something.
- 💡: We can treat $A \cap B$ (or any union or intersection) as *one* event!
- 💡: All conditional probabilities *are* probabilities, and all probabilities *are* conditional probabilities (implicitly conditioning on S).
- 💡: Often, we want to find $P(A | B)$ but are given $P(B | A)$. This is where Bayes' Rule is useful!
- 💡: LOTP is useful when we want to condition on something (i.e., “wishful thinking”). Also, LOTP is useful to find the denominator in Bayes' Rule.

7 Exercises

✍️ **Problem 1.** Suppose that we're on the game show *Let's Make a Deal* hosted by Monty Hall. We choose 1 of the 3 closed doors, 2 of which have a goat behind them and 1 of which has a car. Monty, who knows where the car is, then opens 1 of the 2 remaining doors. The

door he opens always has a goat behind it (i.e., he never reveals the car). If he has a choice, he picks a door at random with equal probabilities. Monty then offers us the option of switching to the other unopened door. If our goal is to get the car, should we switch doors?²

- (a) Find the probability that we win the car if we don't switch doors.
- (b) Find the probability that we win the car if we switch doors.
- (c) Now suppose that Monty is a very old man and his memory isn't as good as it used to be. With probability p , he remembers where the car is, and the game proceeds as usual. But with probability $q = 1 - p$, he forgets where the car is, in which case he chooses randomly, with equal probabilities, which door (other than our initial choice) to open. It's possible that he reveals the car, which would spoil the game. If the game is spoiled, Monty apologizes, and we get nothing. Find the probability that we win the car if we switch doors.

Solution

- (a) It's random which of the 3 doors has the car, so by the naive definition of probability, our initial guess has a success probability of $\frac{1}{3}$.
- (b) Let's label the doors 1 through 3. Without loss of generality, assume we picked door 1. (If you don't want to pick door 1, we could simply relabel the doors.) Let A be the event that we win the car, and let C_i be the event that the car is behind door i . By LOTP, $P(A) = P(A | C_1) \times \frac{1}{3} + P(A | C_2) \times \frac{1}{3} + P(A | C_3) \times \frac{1}{3}$. If the car is behind door 1, switching will fail. If the car is behind door 2 or door 3, switching will succeed (since the remaining unopened door must contain the car if Monty always reveals a goat). Thus, $P(A) = 0 \times \frac{1}{3} + 1 \times \frac{1}{3} + 1 \times \frac{1}{3} = \frac{2}{3}$. We should switch doors!
- (c) We'll use the same notation as above. Let R be the event that Monty remembers where the car is. By LOTP, $P(A) = P(A | R)P(R) + P(A | R^c)P(R^c)$. Given R , the game proceeds as usual, so $P(A | R) = \frac{2}{3}$. But given R^c , we win the car if and only if the car is not behind our initial guess (like before) AND Monty doesn't spoil the game (which he would do with probability $\frac{1}{2}$). Thus, $P(A) = \frac{2}{3} \times p + \left(\frac{2}{3}\right) \left(\frac{1}{2}\right) \times q = \frac{p+1}{3}$. Notice that if $p = 1$ (i.e., he never forgets), this reduces to our previous answer.

 **Problem 2.** There are 2 different diseases that cause a certain weird symptom, and anyone who has either or both of these diseases will experience the symptom. Let D_1 be the event of having the first disease, D_2 be the event of having the second disease, and W be the event of having the weird symptom. Suppose that D_1 and D_2 are independent, with $P(D_j) = p_j$, and a person with neither of these diseases will have the weird symptom with probability w_0 . Let $q_j = 1 - p_j$, and assume that $0 < p_j < 1$.³

- (a) Find $P(W)$.
- (b) Find $P(D_1 | W)$ and $P(D_1, D_2 | W)$.

²Inspired by Example 2.7.1 in *Introduction to Probability* by Joseph K. Blitzstein and Jessica Hwang and Problem 1 in "Stat 110 Homework 3, Fall 2024" by Joseph K. Blitzstein.

³Inspired by Problem 1 in "Stat 110 Final from 2016" by Joseph K. Blitzstein.

Solution

(a) Let $D = D_1 \cup D_2$ (i.e., the event of having either of these diseases).

$$\begin{aligned}
 P(W) &= P(W \mid D)P(D) + P(W \mid D^c)P(D^c) && \text{by LOTP} \\
 &= 1 \times P(D) + w_0 \times P(D^c) && \text{by substituting} \\
 &= P(D_1 \cup D_2) + w_0 \times P(D_1^c, D_2^c) && \text{by De Morgan's Laws} \\
 &= P(D_1) + P(D_2) - P(D_1, D_2) + w_0 \times P(D_1^c)P(D_2^c) && \text{by PIE and independence} \\
 &= p_1 + p_2 - p_1p_2 + w_0 \times q_1q_2 && \text{by substituting} \\
 &= 1 - (1 - w_0)q_1q_2 && \text{by simplifying}
 \end{aligned}$$

(b) We'll use the same notation as above.

$$\begin{aligned}
 P(D_1 \mid W) &= \frac{P(W \mid D_1)P(D_1)}{P(W)} && \text{by Bayes' Rule} \\
 &= \frac{1 \times p_1}{P(W)} && \text{by substituting} \\
 &= \frac{p_1}{1 - (1 - w_0)q_1q_2} && \text{from (a)} \\
 P(D_1, D_2 \mid W) &= \frac{P(W \mid D_1, D_2)P(D_1, D_2)}{P(W)} && \text{by Bayes' Rule} \\
 &= \frac{P(W \mid D_1, D_2)P(D_1)P(D_2)}{P(W)} && \text{by independence} \\
 &= \frac{1 \times p_1p_2}{P(W)} && \text{by substituting} \\
 &= \frac{p_1p_2}{1 - (1 - w_0)q_1q_2} && \text{from (a)}
 \end{aligned}$$

 **Problem 3.** Let r be the probability that a student in a class reads the textbook, with $0 < r < 1$. Suppose that solving a certain problem correctly is twice as likely for someone who has read the textbook compared to someone who has not.⁴

(a) Find the probability that a student read the textbook, given they solved the problem correctly.

Solution

Let C be the event that someone solves the problem correctly and R be the event that someone reads the textbook. Let $x = P(C \mid R) = 2P(C \mid R^c)$.

⁴Inspired by Problem 3 in “Stat 110 Homework 2, Fall 2024” by Joseph K. Blitzstein.

$$\begin{aligned} P(R \mid C) &= \frac{P(C \mid R)P(R)}{P(C)} && \text{by Bayes' Rule} \\ &= \frac{P(C \mid R)P(R)}{P(C \mid R)P(R) + P(C \mid R^c)P(R^c)} && \text{by LOTP} \\ &= \frac{x \times r}{x \times r + \frac{x}{2} \times (1 - r)} && \text{by substituting} \\ &= \frac{2r}{r + 1} && \text{by simplifying} \end{aligned}$$